1 · CONVERSIONS

From	To	Note
$a^T x \le b$	$+s = b$	**slack** — is a basis column
$a^T x \ge b$	$-s = b$	**surplus** — *not* a basis column
$\ge b, b<0$	$\times(-1)$ first	then add a slack
$a^T x = b$	$+w = b$	**artificial** — start basis only

2 · READING A TABLEAU

- **Optimal?** every Row-0 entry ≥ 0.
- **Basic variables + values:** Basis column ↔ RHS column, row by row.
- **Everything else = 0.** Say so explicitly.
- **Objective:** Row 0's RHS entry.
- Row 0: under a **basic** column always 0; under **slack** s_i it is the shadow price y_i.

```
c'_j = yᵀa_j − c_j          y = c_B B⁻¹
```

3 · SENSITIVITY

Shadow price y_i — the value of one more unit of resource i. Raise b_i by 1, the objective improves by y_i.

- It is a derivative: $z = y^T b$, so $\partial z / \partial b_i = y_i$.
- Economically: the most you'd pay for one extra unit.
- Same object as: dual variable · simplex multiplier · opportunity cost · π.

RHS ranging

Perturbation — replace b_i by $b_i + \delta$ and ask how far δ can move.

Shifting b_i moves every basic value by $\delta \times$ the entry in the **s_i column** (that column holds B^{-1}).

```
one inequality per basic row:
    RHS_row  +  δ · (sᵢ column entry)  ≥  0
```

- Solve each for δ, intersect → the range where the basis stays optimal.
- Uses only B^{-1} and b — **never c**. Zero prices change nothing.

Degeneracy

Degenerate BFS — a **basic** variable equal to **0** in the RHS column.

- **The two endpoints of the RHS range are exactly the degenerate points.** One table answers both questions.
- The row whose inequality **binds** names the variable that hits 0 — and it is the one that would **leave** the basis.

Past the range

That variable **leaves**. Entering one by the **dual ratio test**: in the leaving row, among entries < 0, minimise $|c'_j / \text{row}_j|$. Then one pivot.

Cost ranging: for a basic variable, use its row under the non-basic columns and force every resulting Row-0 entry ≥ 0.

4 · DUALITY

Deriving the dual

- max ↔ min.
- One dual **variable** per primal **constraint**.
- One dual **constraint** per primal **variable**.
- Dual constraint j reads **down column j** of A.
- Swap c and b.

SOB — classify, then translate

	S sensible	O odd	B bizarre
variable	$x \ge 0$	free	$x \le 0$
constr. in max	$\le$	$=$	$\ge$
constr. in min	$\ge$	$=$	$\le$
constr → dual var	$y \ge 0$	y free	$y \le 0$
var → dual constr	sensible dir.	$=$	bizarre dir.

Sensible direction = $\ge$ if the dual is a min, $\le$ if it's a max.

Weak & strong duality

Weak — $c^T x \le b^T y$ for **all** feasible x and y. Every feasible dual is a ceiling on every feasible primal; they never cross.

Strong — $c^T x^* = b^T y^*$ at optimality. The ceiling is actually reached; no duality gap.

- **Find x, y with $c^T x = b^T y$ ⟹ both are optimal.** One line.
- **"Show (P) is not unbounded"** ⟹ exhibit **one** feasible dual point and check it; weak duality then caps (P). Squeeze both ways to pin z^*.

If the primal is	then the dual is
unbounded	infeasible
infeasible	unbounded **or** infeasible
feasible, and D too	both finite

Complementary slackness

CS — each product $(\text{slack})\cdot(\text{variable}) = 0$, so at most one of the pair is non-zero.

```
primal    ((Aᵀy)_j − c_j) · x_j = 0    per variable
dual      ((Ax)_i − b_i) · y_i = 0     per constraint
```

Write every line out — one per constraint, one per variable. It is separately scored.

Using it: is a given x* optimal?

- **1.** Check x^* is primal **feasible** first.
- **2.** Constraint i has slack ⟹ $y_i = 0$.
- **3.** $x_j \ne 0$ ⟹ dual constraint j holds with =. Solve for y.
- **4.** Test y against the **remaining** dual constraints **and the sign restrictions**.

Step 4 result	Conclusion
all satisfied	x^* **is** optimal; y is the dual optimum
any violation	x^* is **not** optimal

Both verdicts have appeared. Don't assume "not optimal".

5 · IP MODELLING

The answer template

```
z ∈ {0,1}:   z = 1 ⟺ <meaning, full sentence>
     <linking constraint>
     <requirement on z>  ∀i ∈ <explicit range>
```

Four things score **separately**:

- the meaning **in words**
- the **linking** constraint
- the ∀ with its explicit range
- the **domain line** ($z \in \{0,1\}$)

Patterns

Warehouses $i \in I$ serve customers $j \in J$; y_i = built, x_{ij} = serves.

English	Constraint
exactly one	$\Sigma_i\, x_{ij} = 1\ \forall j$
at most k	$\Sigma_i\, y_i \le k$
capacity	$\Sigma_j\, x_{ij} \le c_i\ \forall i$
only if built	$x_{ij} \le y_i\ \forall i,j$ — **strong** weak: $\Sigma_j\, x_{ij} \le \lvert J\rvert \cdot y_i$
$A \Rightarrow B$	$y_A \le y_B$
several premises	$\Sigma(\text{prem}) - (\#\text{prem}-1) \le \Sigma(\text{concl})$
at most one of two	$y_1 + y_2 \le 1$
product $y_1 \cdot y_2 \to Y$	$Y \le y_1, Y \le y_2, Y \ge y_1+y_2-1$
either–or	$f_1 \le b_1+Mz, f_2 \le b_2+M(1-z)$
fixed charge	$x \le M\cdot y, x \ge q\cdot y$
rolling window	$\Sigma_{t=k}^{k+6} x_t \le 5\ \forall k \in \{1..T-6\}$
start-up	$s_t \ge x_t - x_{t-1}, t \ge 2$
ratio / %	clear the denominator, all on one side

Product rule: Y rewarded → first two suffice; penalised → the third; all three is always safe.

Indicator — $t = 1 \iff X > K$, X integer

```
(A)  X ≤ K + M·t       X>K ⟹ t=1    forces t UP
(B)  X ≥ (K+1)·t       t=1 ⟹ X>K    forces t DOWN
```

- "indicates whether" / "if and only if" → write **both**.
- Unsure → write both. Never penalised.
- M is a **constant**. Justify its size: $|J|$, c_i, Σw_j.

TRAPS

- missing ∀, or the wrong range — $J\setminus\{21\}$, $t \in \{4..15\}$
- an auxiliary with no meaning in words, or no domain line
- M treated as a variable
- one direction when the wording said "iff"
- a product left non-linear

6 · BRANCH & BOUND

Write MAX or MIN at the top of your answer.

	relaxation gives	integer c, x
max	an **upper** bound	$\text{OPT} \le \lfloor z_{LP} \rfloor$
min	a **lower** bound	$\text{OPT} \ge \lceil z_{LP} \rceil$

Branching: on a fractional $x_i = f$, split into $x_i \le \lfloor f \rfloor$ and $x_i \ge \lceil f \rceil$. No integer point is lost. Children **inherit** every ancestor constraint.

Closing a node — three rules

Rule	Condition
integrality	LP solution already integral → update Z^* if better, then close
infeasibility	the relaxation has no feasible point
bound	$Z_{node} \le Z^*$ (max) · $Z_{node} \ge Z^*$ (min)

- A **tie** ($Z_{node} = Z^*$) still prunes.
- Integrality closes the node **even when it improves** — there's nothing left to split.
- Record per node: **node · constraint added · vertex · Z · which rule closed it**.

Stopping the algorithm

- Formally: the **open list is empty** → the incumbent is optimal. Never an incumbent → the IP is **infeasible**.
- **Early stop: as soon as every open node's bound cannot beat Z^*.**
- **Justify it in a sentence** — "P_1 is integer with 14, above P_2's 12.9, so above any child of P_2."

Node order

- **FIFO** — queue, breadth-first, take the **oldest** open node.
- **LIFO** — stack, depth-first, take the **newest**.
- **State which child you push first** — the answer depends on it.

On a plot: empty region → prune by infeasibility, no arithmetic. Otherwise slide the objective line to the last vertex it touches.

7 · TOTAL UNIMODULARITY

TU — every **square submatrix** has determinant in $\{-1, 0, +1\}$. Hence every entry is too.

TU + integral b ⟹ integral polyhedron ⟹ the IP solves in polynomial time.

Proving it

- **By recognition** — the incidence matrix of a **bipartite** or **directed** graph is TU. Usually the whole answer.
- Consecutive-ones (interval) matrices are TU.
- **By the three conditions:**
  - entries in $\{-1,0,+1\}$
  - at most 2 non-zeros per **column**
  - rows split into M_1, M_2: same sign → different parts, opposite signs → same part

Do not write "Ghouila-Houri" — reproduce the three conditions.

Disproving it

Find a **zero-free 2×2** with $|\det| \ge 2$. Any 2×2 containing a zero has $\det \in \{-1,0,1\}$, so only zero-free blocks can break it.

Closure: A TU ⟹ $-A$, A^T, A^{-1}, $[A, I]$ are all TU.

8 · MATROIDS

Greedy finds the optimum for every weight function ⟺ the system is a matroid.

E — the ground set, a finite pile.
$I \subseteq 2^E$ — the independent sets: which selections are allowed.

The three axioms

- **(1)** $\varnothing \in I$
- **(2)** $B \in I$ and $A \subseteq B \Rightarrow A \in I$ *(hereditary)*
- **(3)** $A, B \in I$ with $|A| < |B| \Rightarrow \exists x \in B\setminus A$ with $A \cup \{x\} \in I$

(1)+(2) alone = an **independence system**. Adding (3) makes it a matroid.

Basis

Basis — a **MAXIMAL** independent set: it is in I and cannot be extended.

- bases $\subseteq I$. A basis is not extra structure — it's a member of I.
- **Maximal ≠ maximum.** Maximal = cannot be extended (**the definition**). Maximum = largest possible size (**the theorem**).
- Writing "the largest independent set" as the definition **loses the mark**.

All bases of a matroid have the same size — so here maximal ⟹ maximum.

To disprove: exhibit $A, B \in I$ with $|A| < |B|$ such that **nothing** in $B\setminus A$ extends A. Three elements is usually enough.

9 · KNAPSACK DP

$B[i,w]$ — the best value using **only items 1..i** with capacity w.

Row 0 and column 0 are all **zero**: no items, or no room, means no value.

Recurrence — one decision per cell

```
if w_i ≤ w:
   B[i,w] = max{ B[i−1, w] ,              ← SKIP i
                 v_i + B[i−1, w−w_i] } ← TAKE i

if w_i > w:
   B[i,w] = B[i−1, w]          doesn't fit
```

Backtracking from (n, W)

- $B[i,w] = B[i-1,w]$ → item i **not** taken → go to $(i-1, w)$
- otherwise → item i **was** taken → go to $(i-1, w-w_i)$

Complexity

Index by	Cost
weight	$O(n \cdot W_{max})$
value	$O(n \cdot V_{max})$

Run whichever bound is smaller, and say why. This is a marked step.

Pseudopolynomial, not polynomial: W is a *number*, written in $\log W$ bits, so $O(nW)$ is exponential in the input *length*.

FPTAS

```
θ = ε·v_max/n ;   v_i* = ⌊v_i/θ⌋
run the VALUE-indexed DP on the scaled data,
report the chosen items at ORIGINAL values
V_approx ≥ (1−ε)·V_opt
P ⊆ FPTAS ⊆ PTAS ⊆ APX
```

Max clique, set cover, max independent set are **not in APX at all**.

10 · NETWORK FLOW

Flow — a number on every arc obeying:
capacity $0 \le f(e) \le u(e)$
conservation in = out at every node except s and t

$val(f) = \Sigma_j f(s,j)$ — the total leaving s, equal to the total arriving at t.

Cuts

Cut $S = [X, V \setminus X]$ with $s \in X, t \notin X$.
$cap(S) = \Sigma u(i,j)$ over $i \in X, j \notin X$ — **forward arcs only**.

Why forward only: capacity measures what could cross toward t. An arc pointing the wrong way carries nothing that direction, so it imposes no limit. Counting it inflates the cut.

```
val(f) ≤ cap(S)   every flow, every cut    ← weak
max val(f) = min cap(S)                    ← strong
```

Find a flow and a cut with the same number ⟹ both optimal. One line.

Residual network

- **Forward** arc (i,j): capacity $u(e) - f(e)$ — spare room.
- **Backward** arc (j,i): capacity $f(e)$ — flow you could **undo**.
- Drop arcs of residual capacity 0.

The backward arc is the **undo button** — without it greedy path-finding is not correct.

Ford–Fulkerson

- **1.** Build the residual network.
- **2.** Find **any** s–t path. None left → the flow is maximal, stop.
- **3.** κ = smallest residual capacity on the path (the bottleneck).
- **4.** f += κ on forward arcs, f −= κ on backward arcs. Repeat.

Edmonds–Karp = the same, always taking a **shortest** path (BFS).
FF: $O(|E| \cdot U)$ pseudopolynomial · EK: $O(V \cdot E^2)$ polynomial.

Reading the min cut for free

- Take the **final** residual network (no augmenting path left).
- X = every node **reachable from s**.
- $S = [X, V \setminus X]$ is a minimum cut. **Check $cap(S) = val(f)$.**

Modelling transformations

many sources/sinks → super-source and super-sink · **node** capacity → split $v\_in \to v\_out$ joined by an arc of that capacity · undirected edge → two opposite arcs

11 · TSP

EULER = every EDGE once (easy). HAMILTON = every NODE once (NP-complete).

TSP = a minimum-weight Hamiltonian cycle.

- **symmetric** $c_{ij} = c_{ji}$ → $(n-1)!/2$ tours
- **asymmetric** → $(n-1)!$ tours
- **metric** $c_{ik} \le c_{ij} + c_{jk}$ — both ratios below need it

The IP, in two stages

Degree constraints: $\Sigma_{i\neq j} x_{ij} = 1\ \forall j$ and $\Sigma_{j\neq i} x_{ij} = 1\ \forall i$.

These alone are just the **assignment problem** — they permit **subtours**. So add:

	Constraint	Count	Relax.
SEC	$\Sigma_{i,j\in U} x_{ij} \le \|U\|-1$, $2\le\|U\|\le n-1$	exp.	**tight**
MTZ	$u_i - u_j + 1 \le (n-1)(1-x_{ij})$, $u_1=1$, $2\le u_i \le n$	$O(n^2)$	**weak**

Use the lecture's $(n-1)$, not the textbook's n.

Approximation

	Ratio	How degrees are made even
MST-doubling	**2**	**double every edge**
Christofides	**3/2**	**min-weight matching on the ODD-degree vertices**
nearest nb.	none	—

- Both: MST → make degrees even → Euler tour → shortcut.
- A better ratio is a **worst-case guarantee**, not a promise of a better answer on an instance.
- The papers **mislabel MST-doubling as "Christofides"**. Execute what's described, add one line naming it correctly.

Classifying a problem

pair one-to-one → **assignment** · one budget, max value → **knapsack** · fewest containers → **bin packing** · cover every element → **set covering** · visit every node → **TSP** · pick nodes so every **edge** is touched → **vertex cover**

⚑ COUNTEREXAMPLE — 7× IN 3 PAPERS

Asked in **matroids · flow · TSP · convexity**. Always the same four moves:

- **1.** Build the **smallest** instance meeting **every** stated assumption — and check them explicitly.
- **2.** State what the claim predicts.
- **3.** Show what actually happens.
- **4.** Write: "*therefore the statement is false in general.*"

Levers that worked:

- **matroid** — $A, B \in I$, $|A|<|B|$, nothing in $B \setminus A$ extends A
- **heuristic** — greedy commits to a cheap early edge and pays later
- **convexity** — make K **non-convex**: $\min x^2$ s.t. $x^2 \ge 1$ gives $x = \pm1$, so the minimum isn't unique
- **flow** — the graph below breaks most capacity claims

```
s→a=1, s→b=1, a→c=1, b→c=1, c→t=3

max flow = 2   — EVEN, though every capacity is odd
min cut ({s},rest) = 2
add λ=2 to every capacity:
  ({t},rest) = 5  beats the old cut's 6
  ⟹ the minimum cut MOVED
```

12 · NONLINEAR, UNCONSTRAINED

Critical point — a point where $\nabla f = 0$. Maxima, minima **and saddles** all qualify, which is why you need the Hessian.

```
∇f = (∂f/∂x₁, …, ∂f/∂xₙ)ᵀ
H_f = (∂²f/∂xᵢ∂xⱼ)    — always symmetric
```

Procedure

- Solve $\nabla f = 0$ for **all** critical points — **factor, don't divide**. $x \cdot (\dots) = 0$ has two branches.
- Compute H_f symbolically.
- **Substitute each point separately** — different points can classify differently.

Test A · 2×2 · $\det = ad - b^2$, $tr = a + d$

det	tr	The point is a
> 0	> 0	positive definite → **minimum**
> 0	< 0	negative definite → **maximum**
< 0	—	indefinite → **saddle**
$= 0$	—	inconclusive → Test C

Why: $\lambda_1\lambda_2 = \det$, $\lambda_1+\lambda_2 = tr$. Same sign, or opposite.

Test B · leading principal minors

- all $D\_k > 0$ → positive definite → **minimum**
- $D_1<0, D_2>0, D_3<0, \dots$ alternating, **starting negative** → **maximum**
- no $D\_k = 0$ but neither pattern → **saddle**
- any $D\_k = 0$ → the test is **inapplicable**

Test C · eigenvalues — the fallback

Solve $\det(H - \lambda I) = 0$. All + → min · all − → max · mixed → saddle · some 0 → truly inconclusive.

A zero minor does NOT mean semidefinite — it means the test can't decide.

Do this FIRST

If $H > 0$ **everywhere** (not just at the critical point) then f is strictly convex, so:

- **no maxima exist at all** — a scoring statement, not a remark
- any critical point is the **unique global minimum**

13 · CONVEXITY

convex objective + convex feasible set ⟹ every LOCAL minimum is GLOBAL.

Convex SET — $\lambda x + (1-\lambda)y \in C$ for all $x, y \in C$, $\lambda \in [0,1]$.
The whole segment between any two points stays inside.

Convex FUNCTION — $f(\lambda x+(1-\lambda)y) \le \lambda f(x)+(1-\lambda)f(y)$.
The chord between two points on the graph lies above the graph.

- **Strictly convex** — < for $x \neq y$. x^2 is; $|x|$ is convex but not strictly.
- **Concave** — reverse the inequality; equivalently $-f$ is convex.
- **Affine $a^\top x + b$** — **both** convex and concave (the chord *is* the graph).

	compact	convex
circle **curve** $x^2+y^2=r^2$	✓	✗
closed **disk** $x^2+y^2\le r^2$	✓	✓
halfplane	✗	✓
intersection of convex		✓ always
union of convex		✗ generally

Open / closed / bounded / compact

- **Open** — every point has a small ball still inside. Strict <.
- **Closed** — contains its own boundary; complement is open. Non-strict ≤.
- **Bounded** — fits inside a finite box.
- **COMPACT = closed AND bounded** (Heine–Borel, in $\mathbb{R}^n$).
- "Not open" ≠ "closed": $[0,1)$ is neither, $\mathbb{R}^n$ is both.
- On $(0,1)$ the max is never attained — not closed. On $[0,\infty)$ it runs away — not bounded.

Weierstrass: non-empty compact set + continuous f ⟹ a global min AND max are attained.

Proving convexity

- **From the definition** — when f is abstract or non-differentiable (a norm).
- **Hessian** — on a **convex domain**: $H \ge 0$ everywhere ⟺ convex; $H > 0$ everywhere ⟹ strictly convex (converse false — x^4).
- **Composition** — preserved by αf ($\alpha\ge0$), $f+g$, $f(Ax+b)$, $\max\{f_1..f_m\}$, $h\circ g$ (h convex, non-decreasing).
- **Counterexample** — one triple (x,y,λ) disproves it.

The norm proof — say every line

```
‖λx+(1−λ)y‖ ≤ ‖λx‖ + ‖(1−λ)y‖
                    triangle inequality
            = |λ|·‖x‖ + |1−λ|·‖y‖
                    homogeneity
            = λ‖x‖ + (1−λ)‖y‖
                 λ∈[0,1] ⟹ both ≥ 0   ←SAY THIS
```

14 · KKT

Standard form — convert first, always

```
min f(x)    s.t.  g_i(x) ≤ 0 ,  h_j(x) = 0

g(x) ≥ 0   becomes   −g(x) ≤ 0
max f      becomes   min −f
```

- **ACTIVE** (tight, binding): $g_i(x^*) = 0$ — the constraint presses.
- **INACTIVE** (slack): $g_i(x^*) < 0$ — room to spare.

The Lagrangian

```
L = f(x) + Σᵢ λᵢ gᵢ(x) + Σⱼ μⱼ hⱼ(x)      (MIN)

λᵢ  inequality  −  λᵢ ≥ 0  SIGN-CONSTRAINED
μⱼ  equality    −  free in sign
```

Why: violating $g_i \le 0$ must *increase* the objective you're minimising, so λ can't be negative. An equality breaks either way, so μ needs both signs. Same SOB pattern as §4. **λ is a shadow price.**

The four blocks

stationarity	$\nabla f + \Sigma\lambda_i\nabla g_i + \Sigma\mu_j\nabla h_j = 0$
primal feasibility	$g_i \le 0$, $h_j = 0$
dual feasibility	$\lambda_i \ge 0$
compl. slackness	$\lambda_i \cdot g_i(x^*) = 0$

Solving — the case split

For each i: **either** $\lambda_i = 0$ (inactive) **or** $g_i(x^*) = 0$ (active).

- Discard a case if $\lambda_i < 0$ — dual infeasible.
- Discard a case if $g_i > 0$ — primal infeasible.
- **Say why each branch died.** It's scored.

Kill cases early using a fact in the problem statement — *two distinct points exist* $\Rightarrow r > 0 \Rightarrow \mu r = 0 \Rightarrow \mu = 0$.

Slater vs LICQ — this decides your conclusion

	Requires	Gives you
Slater	f, g_i convex; h_j affine; a point $\bar{x}$ with **every $g_i(\bar{x}) < 0$ strictly**	a KKT point **IS a global optimum** — stop
LICQ	active ∇g_i plus all ∇h_j linearly independent	**candidates only** — evaluate f and compare

- **Verify Slater by exhibiting the point.** Centroid with the radius inflated by +1, so every $g_i \le -1 < 0$. The +1 manufactures strictness.
- **Only equality constraints ⟹ Slater always fails** — an equality has no strict version.

MC TRAP

- "a KKT point is always a local optimum" — **FALSE**
- "for a convex problem a KKT point is a global minimiser if Slater holds" — **TRUE**

The four-step recipe

- **1. Existence** — compact + continuous ⟹ a global min and max exist (Weierstrass). Cheap, often forgotten.
- **2. CQ** — Slater? otherwise LICQ.
- **3. Solve** — Lagrangian → 4 blocks → case split → discard, with reasons.
- **4. Compare** — evaluate f at survivors. **Skip if Slater held.**